

Detecting Label Shift for Large Language Models with Conformal Test Martingales

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Abstract

We study the problem of detecting label shift in deployed Large Language Models (LLMs), using model outputs, in the form of class labels, confidence estimates, and justifications, as proxies for true labels. We propose the use of conformal test martingales, which can detect shifts in both mean and dispersion across multiple streams while controlling the false alarm rate. We evaluate the approach on three binary and three multiclass text classification datasets using LLMs ranging from 3B to 32B parameters. We consider three martingale algorithms: Simple Jumper, Sleeper/Stayer, and Sleeper/Drifter, as well as their composite. The observed false alarm rates are consistently below the nominal level. Detection-time rankings are stable across datasets, models, and signals, with Sleeper/Drifter and the composite martingale outperforming the others, while Simple Jumper is least competitive. Larger models tend to enable earlier detection of label shift, although the effect is uneven and non-monotonic. Among the signals, confidence is the most reliable proxy for binary datasets, whereas combining all three signals yields the best performance on two of the multiclass datasets. Justification is highly dataset-dependent: it fails for several datasets and models, yet is the only effective signal for the smallest model in one case.

Keywords: label shift, large language models, conformal test martingales.

1. Introduction

The decision to deploy a predictive model is often based on an evaluation using data that is assumed to be sampled independently from the same underlying distribution as the production data. Violations of this assumption may lead to that deployed models do not meet requirements, e.g., regarding the expected error rate. Moreover, validity guarantees for uncertainty quantification often rely on the assumption of exchangeability, i.e., any order of the observations is equally probable (Vovk et al., 2022), which is violated if the sampling distribution is drifting. In these and many other scenarios, it is crucial to signal as early as possible when a drift has been detected, while controlling the probability of a false alarm.

Concept drift occurs when the joint probability distribution $\mathbb{P}(\mathbf{x}, y)$ varies over time, where $\mathbf{x}$ is an object and y is a label (Gama et al., 2014). By rewriting the joint probability distribution as $\mathbb{P}(\mathbf{x}, y) = \mathbb{P}(y) \mathbb{P}(\mathbf{x}|y)$, we can identify two special cases of concept drift; *label shift* (or class prior shift, in the context of classification), where $\mathbb{P}(y)$ varies over time, and *label-conditional (or class-conditional) covariate shift*, where $\mathbb{P}(\mathbf{x}|y)$ varies over time. While both label and label-conditional covariate shift may occur simultaneously, there are also situations where only one of the two types is present, e.g., label-shift occurs on its own when

the probability of infection increases during a pandemic, while the probability of observing certain symptoms given the infection remains the same.

In many real-world scenarios, test objects are observed over time without any accompanying labels, e.g., adverse drug events may take months to develop. Several unsupervised drift detectors have been proposed to handle such challenging scenarios, many of which rely on comparing a recent and reference window (possibly of varying size) using some data modeling technique, e.g., clustering or probability density estimation, as described in a recent survey (Lukats et al., 2025). In some cases a trained classifier is employed, e.g., using artificial labels assigned to instances in the reference and recent window, respectively (Gözüağık et al., 2019), but few approaches target scenarios in which drift is detected relative to a previously trained model. One exception is (Lipton et al., 2018) in which predicted labels of a given model are used to estimate (and correct) distribution shift, under a few assumptions, such as the confusion matrix being invertible. Another example is (Dos Reis et al., 2016), in which drift detection is studied using either individual features or predicted class probabilities.

In this work, we focus on detecting label shift for deployed Large Language Models (LLMs). These models have been shown to achieve state-of-the-art performance in many different domains and excel in particular when it comes to natural language processing (Zhao et al., 2023; Chang et al., 2024). Such models can be instructed to output not only class labels, but also other information that could potentially be of value for detecting label shift, such as their confidence in the predictions (Xiong et al., 2024). We consider the task of text classification, where an LLM is requested to classify texts into specified categories and also provide a confidence assessment and a justification in the form of a set of keywords. Formally, we are no longer analyzing drift of the probability distribution $\mathbb{P}(y)$, but of $\mathbb{P}(y')$, where y' is the model output, under the assumption that it is correlated with the true label. We will investigate this assumption for the various types of model output.

In order to test if some quantity of interest differs significantly between two samples, e.g., observed before and after a certain point in time, we could employ some suitable statistical test. It should be noted that in an online setting, we would have to employ the test repeatedly, which increases the risk for a type I error (falsely signaling for data drift) due to multiple hypothesis testing. We could apply Bonferroni correction, which is the standard tool for mitigating the problem, if the number of tests is known in advance. This is however typically not the case in an online setting, where the stream may be infinite. The ADWIN algorithm (Bifet and Gavalda, 2007) addresses this problem and provides a guarantee on the false alarm rate, but may only detect shifts in the mean and does not handle multiple streams. Conformal test martingales (Vovk et al., 2022) is a more powerful alternative, which enables the detection of distributional changes in both mean and dispersion, while controlling the probability of a false alarm for both single and multiple streams.

The main contributions of the paper are:

- We show that conformal test martingales can detect label shift with different types of LLM output; predicted labels, confidence assessments and justifications
- We present an extensive empirical investigation, covering binary and multiclass text classification, LLMs of different sizes and multiple algorithms for computing conformal test martingales.

In the next section, we briefly introduce conformal test martingales and propose a way of adapting them to the task of detecting label shift for LLMs. In Section 3, we describe experimental setup and results. Finally, we summarize the main findings and outline some directions for future work in Section 4.

2. Methods

2.1. Conformal Test Martingales

Given a sequence of objects $\mathbf{x}_1, \mathbf{x}_2, \dots$, where each $\mathbf{x}_i \in \mathcal{X}$, for some object space $\mathcal{X}$, for which a sequence of scores $\alpha_1 = A(\mathbf{x}_1), \alpha_2 = A(\mathbf{x}_2), \dots$ is obtained using some scoring function $A : \mathcal{X} \rightarrow \mathbb{R}$. From the sequence of scores $\alpha_1, \alpha_2, \dots$ and a sequence of values $\tau_1, \tau_2, \dots$ that are sampled IID and uniformly from the unit interval, a p-value for the n th score is obtained in the following way:

$$p_n = \frac{|i : \alpha_i > \alpha_n| + \tau_n |i : \alpha_i = \alpha_n|}{n} \quad (1)$$

where $i \in \{1, \dots, n\}$ (Vovk et al., 2022). For each p-value in the sequence, a *conformal test martingale* is computed by $s_n = f_1(p_1) \cdots f_n(p_n)$, where each f_i is a *betting function* $f : [0, 1] \rightarrow [0, \infty]$ satisfying $\int_0^1 f(u) du = 1$. If the objects $\mathbf{x}_1, \mathbf{x}_2, \dots$ are sampled IID, then it follows:

$$\mathbb{P}(\{(\mathbf{x}_1, \mathbf{x}_2, \dots) : \exists n : s_n \geq c\}) \leq 1/c \quad (2)$$

where c is some positive constant (Vovk et al., 2022). The conformal test martingale can hence be used for testing the IID assumption, where the probability of a false alarm is controlled by choosing a suitable c . A common choice for the latter is $c = 100$, implying that the probability of observing a conformal test martingale greater than or equal to this is less than or equal to 1%, if the objects are sampled IID.

In addition to choosing a scoring function (A), which we discuss in the next section, also a betting function and an algorithm for computing the conformal test martingale have to be selected when implementing the framework. In this work, we have chosen the three algorithms described in (Vovk et al., 2022): Simple Jumper, Sleeper/Stayer and Sleeper/Drifter. Simple Jumper employs a set of betting functions of the following form:

$$f_\epsilon(p) = 1 + \epsilon(p - 0.5) \quad (3)$$

where $\epsilon \in [-2, 2]$ is a parameter determining the maximum gain (and loss) and whether a p-value below 0.5 gives a gain or not. Similar to ADWIN (Bifet and Gavaldà, 2007), this betting function allows for detecting shifts in the mean only.

Sleeper/Stayer and Sleeper/Drifter employ betting functions of the form:

$$f_{a,b}(p) = \begin{cases} \frac{b}{a} & \text{if } p \leq a \\ \frac{1-b}{1-a} & \text{otherwise} \end{cases} \quad (4)$$

where the parameter a determines the threshold value (to compare the observed p-value p with) and b determines the gain (if $b > a$) or loss (if $b < a$), if $p \leq a$. The algorithms

combine multiple such betting functions, allowing for detecting shifts not only in the mean but also in the dispersion.

As an alternative to choosing one of the martingales generated by the above algorithms, we also form a *composite martingale* (Vovk et al., 2022) from all of them, i.e., by averaging the individual martingales. The probability of a false alarm remains the same, since the composite martingale is still a conformal test martingale, while this avoids placing all bets on one of the algorithms.

2.2. Scoring Functions

We consider different scoring functions from the model output $\mathbf{M}(\mathbf{x})$, where $\mathbf{M}$ is the selected LLM and $\mathbf{x}$ is a prompt. The first type of scoring function (*label*) consider the output class labels. For binary classification tasks, a single scoring function returning a binary value encoding the class label is used, i.e., $A_{\text{label}}(\mathbf{M}(\mathbf{x})) \in \{0, 1\}$, while for multiclass tasks, one binary scoring function $A_{\text{label},k}$ is employed for each class label k .

The second type of scoring function (*confidence*) exploits the assessed confidence in the predicted class label. Let $\gamma \in [0, 1]$ be the confidence extracted from $\mathbf{M}(\mathbf{x})$ and let K be the number of class labels. To guarantee that the predicted label is assigned the highest confidence, we clip the confidence from below, $\tilde{\gamma} = \max(\gamma, 1/K)$. For binary classification, the score equals $\tilde{\gamma}$ if the binary encoding of the predicted label is 1, and $1 - \tilde{\gamma}$ otherwise, i.e., $A_{\text{conf}}(\mathbf{M}(\mathbf{x})) = l\tilde{\gamma} + (1 - l)(1 - \tilde{\gamma})$, where l is the binary encoding of the predicted label. For multiclass tasks, we again use a separate scoring function for each class label k , defined in the analogous way; $A_{\text{conf},k}(\mathbf{M}(\mathbf{x})) = l_k\tilde{\gamma} + (1 - l_k)\frac{1-\tilde{\gamma}}{K-1}$, where l_k is the binary label output for class label k .¹

The third type of scoring function (*justification*) utilize the justification (expressed as keywords in free text), by computing the cosine similarity of an embedding of the text to the embeddings of a set of reference texts. The scoring function computes the average of the cosine similarity to the k most similar embeddings in the reference set. Instead of relying on a given set of reference texts, the LLM is requested to generate such a set, with instructions tailored to the specific task. It should hence be noted that in the proposed approach, the selected LLM does not utilize any training data apart from what has been used for the pretrained model, i.e., a zero-shot learning scenario is considered. For binary classification, only one set of reference texts (characterizing the positive class) is generated, while one set of texts per class is generated in the multiclassification context.

For multiclass classification, there are hence as many sequences of scores per type of output as there are classes, and a single martingale is formed from each individual sequence. However, to avoid elevating the risk of a false alarm, Bonferroni correction is employed, i.e., given a threshold c and k sequences, the corrected threshold becomes ck . This strategy is also considered when combining the different output types; in addition to using martingales for each output type separately, martingales generated from all output types are used jointly, again with Bonferroni correction.

1. In initial experiments, the LLMs were requested to output class probabilities, rather than a confidence score, which resulted in poor alignment between predicted probabilities and labels.

3. Experimental Investigation

In this section, we present an evaluation of using conformal test martingales for detecting label shift, using various output of LLMs as proxies for the true target labels.

3.1. Experimental Setup

3.1.1. DATASETS

Six datasets for text classification were investigated²; three concern binary classification: Spam Text Message Classification (SPAM, 5 572 texts), IMDB Dataset of 50K Movie Reviews (IMDB, 50 000 texts), and Fake and Real News Dataset (FAKE, 44 898 texts³), and three concern multiclass classification: Twitter Tweets Sentiment Dataset (TWEETS, 27 480 texts, 3 classes), BBC Full Text Document Classification (BBC, 2 225 texts, 5 classes), 20 newsgroups (NEWS, 18 331 texts, 7 classes⁴).

3.1.2. LLMs

Four variants of Qwen 2.5 (Qwen et al., 2025) were employed, ranging from 3B, via 7B and 14B, up to 32B parameters, together with a separate model, nomic-embed-text v. 1.5, to generate text embeddings.⁵ All models were downloaded and executed locally through the open-source platform vLLM v. 0.17.0 (Kwon et al., 2023).

The prompts given to the LLMs provided a brief description of the task together with a request to output the predicted class label, an assessment of the confidence and a justification, followed by the text to be classified; see Appendix A for an example prompt and model output. The prompts resulted in valid output (in JSON format) in all but two cases; once for the 7B model and once for the 32B model on the BBC dataset, corresponding to a failure rate of 0.0004 for the two models on this dataset. The failure rate for these models on all other datasets, as well as for the two other models on all the datasets, was 0.0.

As described in Section 2.2, scoring functions were formed from the extracted labels, confidence assessments and justifications. For the latter, the models were requested to generate sets of reference texts; see Appendix B for an example of the output of the 32B model for the TWEETS classification task. The number of neighbors in each set of reference texts that were used to obtain the justification-based scores was set to $k = 5$.

In Table 1, the area under ROC curve for all scoring functions and the correct labels are shown; for the multiclass datasets, the AUC is computed using the micro-averaged one-vs-rest strategy. The ranking performance varies across datasets and models, where the larger models tend to outperform the smaller ones. It can also be observed that the use of

2. The first five datasets were obtained from <https://www.kaggle.com/datasets/> while the last is from https://scikit-learn.org/stable/datasets/real_world.html.

3. The titles were used for this dataset rather than the text bodies, as the latter revealed class membership by providing locations and news agencies for non-fake news only.

4. The original 20 groups were merged into the following: "comp" from all comp.* groups, "rec.sport" from rec.sport.baseball and rec.sport.hockey, "rec.vehicles" from rec.autos and rec.motorcycles, "religion" from alt.atheism, soc.religion.christian and talk.religion.misc, "sci" from sci.crypt, sci.electronics, sci.med and sci.space, "talk.politics", from talk.politics.guns, talk.politics.mideast and talk.politics.misc, and "misc.forsale" (original).

5. The models were downloaded from Hugging Face (<https://huggingface.co/>) and have the following IDs: Qwen/Qwen2.5-*--Instruct, where * is 3B, 7B, 14B or 32B, and nomic-ai/nomic-embed-text-v1.5.

Table 1: AUC for all signals and LLMs

Signal	Model	SPAM	IMDB	FAKE	TWEETS	BBC	NEWS
label	3B	0.807	0.930	0.722	0.734	0.895	0.604
	7B	0.835	0.932	0.883	0.731	0.919	0.814
	14B	0.867	0.934	0.860	0.740	0.962	0.873
	32B	0.867	0.942	0.931	0.760	0.967	0.889
conf.	3B	0.813	0.974	0.721	0.790	0.934	0.697
	7B	0.908	0.980	0.926	0.773	0.965	0.880
	14B	0.962	0.983	0.955	0.807	0.986	0.920
	32B	0.974	0.984	0.961	0.821	0.984	0.934
just.	3B	0.763	0.853	0.636	0.620	0.946	0.812
	7B	0.846	0.916	0.872	0.681	0.944	0.873
	14B	0.846	0.875	0.828	0.709	0.944	0.884
	32B	0.849	0.906	0.927	0.726	0.951	0.893

confidence generally leads to better rankings than when using the labels, which in turn is followed by using the signals provided by the justifications. Some notable exceptions include the 7B model outperforming both the 14B and 32B model when using justifications on the IMDB dataset, and the 3B model outperforming the 7B model on TWEETS when using either the labels or the confidence.

3.1.3. CONFORMAL TEST MARTINGALES

The three algorithms Simple Jumper (SJ), Sleeper/Stayer (SS) and Sleeper/Drifter (SD) were used together with a composite martingale (CM) formed from these three. The conformal test martingales were implemented using the Python package `crepes` (Boström, 2024), where default parameters for the algorithms and betting functions, as suggested in (Vovk et al., 2022), were used.⁶ Label shift was signaled across all tasks using a 99% confidence level ($c = 100$).

3.1.4. TEST PROTOCOL

For each task, an initial sequence of texts was randomly selected from the complete batch, keeping the relative class frequencies the same as in the original batch. Label shift was emulated by synthetically changing the class priors after the initial sequence, and a second sequence was obtained by sampling according to the new class priors from the remaining batch. The length of the sequences are shown in Table 2, where *Changepoint* indicates when the class prior was changed, *Endpoint* indicates the length of the full sequence, and *Window* gives the number of observations after the changepoint. For the binary classification tasks (SPAM, IMDB and FAKE), the class prior (p) for the positive class after the changepoint was sampled uniformly according to $p \sim \mathcal{U}(0.05, 0.95)$. For the multiclass datasets (TWEETS,

6. See <https://crepes.readthedocs.io/en/latest/crepes.martingales.html> for the documentation of the recently added `crepes.martingales` module.

Table 2: Change- and endpoints for label shift

Dataset	Changepoint	Endpoint	Window
SPAM	100	700	600
IMDB	500	3000	2500
FAKE	500	3000	2500
TWEETS	500	1000	500
BBC	500	1000	500
NEWS	500	1000	500

BBC and NEWS), the class priors ($\mathbf{p}$) of the K classes after the changepoint were sampled from the symmetric Dirichlet distribution with all concentration parameters equal to 1, i.e., $\mathbf{p} \sim \text{Dir}(\mathbf{1}_K)$ where $\mathbf{1}_K$ denotes the K -dimensional vector of ones. The performance of the conformal test martingales were measured by the index at which they signal for a distribution shift, at 99% confidence. If no drift was detected, the signal index was set to the length of the entire sequence, i.e., the endpoint. In addition, the false alarms, i.e., signals prior to the changepoint, were recorded.

3.2. Experimental Results

3.2.1. ILLUSTRATIONS

Before presenting the full results, we first illustrate the effect of a changed class prior on the signaling time, i.e., the index at which a distribution shift is signaled for, using a couple of models and binary datasets.

In Fig. 1, the signaling times for martingales formed from the output of the 3B model on the SPAM dataset are shown, averaged over 100 iterations of the above procedure, while varying the class prior after the change point (indicated by the dotted red line at index 100) from 0.05 (just below the original class prior, indicated by the dashed gray line) up to 0.95. It can be clearly seen that as the class prior increases, the earlier a label shift can be detected. For this dataset and model, the martingale using only confidence allows for the earliest detection on average, while the Bonferroni-corrected martingale formed from all three signal types is not far behind.⁷

In Fig. 2, the signaling times for martingales formed from the 32B model on the FAKE dataset are shown, when again the results were averaged over 100 iterations, while varying the class prior after the change point (at index 500) from 0.05 up to 0.95. As the changed class prior gets more extreme (on the lower or upper side of the original prior which is 0.5), the earlier a label shift can be detected. Also for this dataset and model, the justification-based signal is lagging a bit behind the others.

7. It should be noted that the combined martingale can never signal earlier than the earliest of the constituent martingales for a single run, but can still be the best-performing on average.

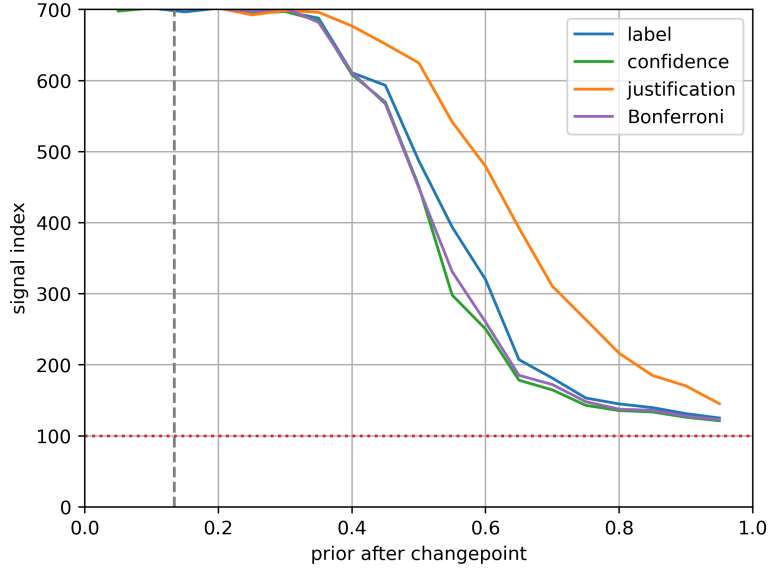


Figure 1: Martingales formed from the 3B model on the SPAM dataset

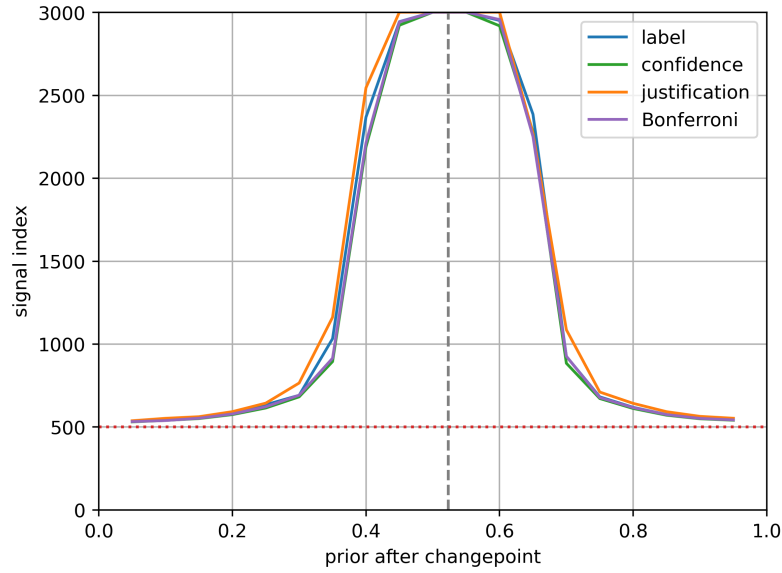


Figure 2: Martingales formed from the 32B model on the FAKE dataset

Table 3: Average false alarm rates for all 64 combinations on all datasets

Dataset	False alarm rate
SPAM	0.0004
IMDB	0.0041
FAKE	0.0038
TWEETS	0.0032
BBC	0.0022
NEWS	0.0016

3.2.2. FULL RESULTS

For the full investigation, the test protocol was repeated 1000 times (with unique random seeds) for each dataset, on all combinations of signals (4), models (4) and conformal test martingales (4), i.e., in total 64 combinations.

Table 3 shows the recorded false alarm rates for each dataset, averaged over all 64 combinations. Reassuringly, the average false alarm rates are lower than the requested (0.01). The dataset with the lowest average false alarm rate (SPAM) is also the one with the earliest changepoint (at index 100 vs. 500 for the others), which is not very surprising as less opportunities are given for generating a high martingale value by chance.

Table 4 shows the average median time points at which drift is detected from the 1000 repeated runs on the SPAM dataset. Of the model and signal combinations, it is only the smallest (3B) model together with the justification-based signal, that do not allow for detecting a shift in label prior (the median index equals the length of the full sequence). It should be noted, however, that the smallest model performs reasonably well for the other signal types (label, confidence and a combination of all three). It can further be observed that Sleeper/Stayer (SS) is quite consistently outperformed by all other martingales, across all models and signals. On the contrary, Sleeper/Drifter (SD) is frequently the best single algorithm, and only occasionally matched or slightly outperformed by the composite martingale (CM). The benefit of combining the three martingales into the latter is hence modest at best. Another observation for this dataset is that justification improves dramatically with model size; from not allowing drift to be detected at all to allowing drift to be detected before half of the observation window after the changepoint. Confidence is the most effective signal overall for 7B, 14B and 32B, consistently outperforming label and justification, with the 32B/SD combination achieving a median of just 143.5, which is not very far from the changepoint (100).

In contrast to the results for the SPAM dataset, CM results in the earliest detection across nearly all models and signals for the IMDB dataset, as shown in Table 5. Justification is by far the least effective signal, with medians well above the others, but still allows for detecting label shift even for the 3B model, which again contrasts to the SPAM dataset. The detection times fall into a quite narrow range for the label, confidence, and composite signals, and the differences between martingale algorithms and models are small. The 3B model is clearly worse than the others when using justification but comparable with respect

Table 4: Average median detection times for SPAM

Model	Signal	SJ	SS	SD	CM
3B	label	518.0	610.5	374.5	367.0
	conf.	600.0	377.0	286.0	301.5
	just.	700.0	700.0	700.0	700.0
	Bonf.	415.5	424.5	317.5	320.5
7B	label	296.5	540.5	284.0	306.0
	conf.	206.5	228.5	190.0	192.0
	just.	297.5	657.5	465.0	310.5
	Bonf.	229.0	264.0	203.5	208.0
14B	label	240.0	282.0	220.0	232.5
	conf.	187.0	172.5	153.5	156.0
	just.	280.0	414.5	296.0	273.0
	Bonf.	198.5	187.5	162.0	171.5
32B	label	243.5	299.5	230.5	228.0
	conf.	180.0	164.0	143.5	148.0
	just.	283.0	571.5	431.0	308.0
	Bonf.	194.0	175.0	153.5	160.0

to the other signals, which aligns with IMDB having high AUC even for the smallest model; see Table 1.

Table 6 shows that the 3B model completely fails on all signals, except for the label, confidence and composite signals when combined with SD, on the FAKE dataset. Justification fails entirely for all four martingales for the 3B model, mirroring SPAM. The jump in performance between the 3B and 7B models is quite dramatic; from non-detection (in at least half of the iterations) to a median detection around 700–800. CM is clearly dominant for model sizes above 3B. Notably, justification with the 14B model is much worse than for the 7B and 32B models, which again is consistent with the observed AUC. Similar to the results on IMDB, justification is the worst-performing signal, consistent with IMDB, with a substantial performance gap. SD and CM are consistently the best martingales across the 7B, 14B, and 32B models, with SS frequently performing the worst, which hence gives the same ranking of the algorithms as for SPAM and IMDB. The 32B model using a composite martingale with confidence is the single best combination at 668.0.

In Table 7, it can be observed that justification fails completely for all models. The use of labels also fails entirely or almost entirely for the 3B, 7B and 14B models, while the 32B model is the only one that is able to detect drift reliably using labels only. This contrasts sharply with the results for previous datasets, where the use of labels are effective even for smaller models. The 7B model is substantially worse than the 3B model when using confidence or a composite of all signals; the 3B detects deviation in these cases more than 100 observations ahead of the 7B model. This is again consistent with the results on AUC. CM dominates for almost all combinations of signals and models where any drift can

Table 5: Average median detection times for IMDB

Model	Signal	SJ	SS	SD	CM
3B	label	687.0	682.0	670.5	667.0
	conf.	658.5	655.5	655.0	642.0
	just.	839.0	893.5	922.5	852.0
	Bonf.	677.0	666.0	665.5	662.0
7B	label	691.0	677.5	680.0	674.5
	conf.	653.5	648.0	640.5	635.5
	just.	716.0	751.5	720.0	702.5
	Bonf.	666.0	662.5	654.0	660.5
14B	label	686.5	672.0	666.0	662.0
	conf.	645.0	647.0	635.0	632.0
	just.	788.0	823.0	829.0	772.5
	Bonf.	665.0	664.0	655.5	655.5
32B	label	675.0	669.0	660.5	654.0
	conf.	648.0	645.0	634.0	629.0
	just.	716.0	726.5	720.5	706.5
	Bonf.	658.0	653.0	645.0	646.5

Table 6: Average median detection times for FAKE

Model	Signal	SJ	SS	SD	CM
3B	label	3000.0	3000.0	2078.0	2548.0
	conf.	3000.0	3000.0	1995.5	2788.5
	just.	3000.0	3000.0	3000.0	3000.0
	Bonf.	3000.0	3000.0	2695.5	3000.0
7B	label	778.0	786.0	761.0	745.0
	conf.	732.5	751.0	737.5	721.5
	just.	838.0	864.0	882.0	823.0
	Bonf.	727.5	763.0	759.5	721.5
14B	label	842.0	801.5	810.5	785.0
	conf.	694.5	696.0	680.5	677.0
	just.	1000.0	975.5	1032.0	945.5
	Bonf.	719.0	731.0	708.0	712.5
32B	label	719.5	706.5	687.5	684.5
	conf.	685.0	693.5	672.0	668.0
	just.	722.5	755.5	750.0	721.0
	Bonf.	689.5	701.5	688.0	681.0

Table 7: Average median detection times for TWEETS

Model	Signal	SJ	SS	SD	CM
3B	label	1000.0	1000.0	1000.0	1000.0
	conf.	827.5	867.5	838.0	814.0
	just.	1000.0	1000.0	1000.0	1000.0
	Bonf.	895.5	915.0	926.0	878.0
7B	label	1000.0	1000.0	1000.0	1000.0
	conf.	1000.0	984.0	1000.0	928.5
	just.	1000.0	1000.0	1000.0	1000.0
	Bonf.	1000.0	1000.0	1000.0	966.5
14B	label	1000.0	992.5	1000.0	1000.0
	conf.	803.5	835.0	842.5	799.5
	just.	1000.0	1000.0	1000.0	1000.0
	Bonf.	841.0	884.0	901.5	834.5
32B	label	1000.0	904.0	904.5	867.0
	conf.	760.0	801.5	801.0	754.5
	just.	1000.0	1000.0	1000.0	1000.0
	Bonf.	790.0	835.5	836.5	782.0

be detected at all, i.e., when using confidence for all models or label for the largest model. Confidence is the only consistently viable signal for this dataset.

As can be seen in Table 8, SJ is clearly and consistently lagging behind all other martingale algorithms for the BBC dataset. The difference between the best performing algorithms, SD and CM, is often very small, suggesting the extra complexity of forming a composite of the three single algorithms brings negligible benefit here, unlike for IMDB, FAKE and TWEETS. The combined signal is quite consistently faster than confidence alone, and confidence is notably more effective than label, which in turn is on par with justification. Model size has modest but consistent effect; detection times decrease steadily from 3B to 32B, but the differences are quite small.

For the NEWS dataset, the 3B model fails completely when using the label and confidence signals, as can be seen in Table 9. Only the justification-based and combined signals are effective for the 3B model, which is in stark contrast to TWEETS where justification failed completely. SJ is clearly the worst performing algorithm, with a substantial gap to the other algorithms, while no clear winner can be declared among the two best-performing algorithms, SD and CM. The jump in performance from the 3B to the 7B model is quite dramatic when using the label and confidence signals; from non-detection to detecting well before half of the observation window.

In Table 10, the mean rank of martingale and signal combinations, when averaging over all 6000 observations (1000 observations for each dataset) are shown per model, where the combinations are sorted according to the average rank across all models. When comparing 16 methods over 6000 observations at a 95% confidence level ($\alpha = 0.05$), the critical difference (CD), as used for testing whether differences in average ranks are statistically significant for

Table 8: Average median detection times for BBC

Model	Signal	SJ	SS	SD	CM
3B	label	744.5	665.5	652.0	656.0
	conf.	684.0	646.5	638.0	640.5
	just.	665.0	643.0	636.0	635.0
	Bonf.	656.0	632.0	616.5	623.0
7B	label	712.0	647.5	636.0	637.5
	conf.	645.0	626.0	615.0	615.5
	just.	669.0	648.0	636.0	635.5
	Bonf.	640.5	622.5	608.5	612.0
14B	label	679.0	630.0	624.0	624.0
	conf.	649.0	617.5	612.5	615.0
	just.	666.0	645.0	639.0	638.5
	Bonf.	641.0	616.0	604.0	606.5
32B	label	672.5	622.0	617.5	619.0
	conf.	646.5	621.0	610.0	612.0
	just.	660.0	631.0	625.0	623.5
	Bonf.	639.0	613.0	604.0	605.0

Table 9: Average median detection times for NEWS

Model	Signal	SJ	SS	SD	CM
3B	label	1000.0	1000.0	1000.0	1000.0
	conf.	1000.0	1000.0	1000.0	1000.0
	just.	838.5	819.5	817.0	775.0
	Bonf.	791.5	739.0	741.0	721.5
7B	label	884.0	715.5	707.0	706.0
	conf.	694.0	664.5	660.0	654.0
	just.	717.0	672.5	657.0	656.0
	Bonf.	675.0	654.0	644.0	641.0
14B	label	749.5	644.5	623.0	631.0
	conf.	663.0	619.0	611.0	612.0
	just.	699.0	659.0	655.0	654.5
	Bonf.	660.5	622.0	615.0	616.0
32B	label	754.0	650.5	628.5	635.5
	conf.	688.0	632.5	618.0	624.0
	just.	692.0	660.0	645.5	642.0
	Bonf.	668.0	630.5	621.0	622.0

Table 10: Mean rank per model (CD = 0.2978)

Method	3B	7B	14B	32B
SD (conf.)	6.207	5.724	5.066	5.326
CM (conf.)	6.394	5.923	5.379	5.718
SD (Bonf.)	6.813	6.735	6.434	6.449
SS (conf.)	7.257	7.236	6.601	7.080
CM (Bonf.)	7.199	7.142	6.930	7.038
SD (label)	7.949	7.939	7.649	7.342
SS (Bonf.)	7.820	8.216	7.912	8.099
CM (label)	8.283	8.512	8.208	8.031
SJ (conf.)	9.075	8.883	8.777	9.494
SS (label)	8.901	9.333	8.988	9.049
SD (just.)	9.885	9.155	10.152	9.413
SJ (Bonf.)	9.267	9.652	9.730	10.110
CM (just.)	9.716	9.228	10.291	9.601
SS (just.)	10.355	10.149	11.083	10.498
SJ (just.)	10.428	10.778	11.504	11.279
SJ (label)	10.453	11.397	11.296	11.472

any pair of methods using the Nemenyi test, is 0.2978; for any pairs of methods for which the difference in average ranks are larger than this, the corresponding null hypothesis that the methods are equally effective can be rejected. To highlight the significant differences, CD plots are provided for the models in Figs. 3-6, where the null hypothesis for any two methods that are not connected by a vertical line can be rejected. Across all four models, the same two top-performing combinations can be observed, using either Sleeper/Drifter or the composite martingale together with confidence-based signals.

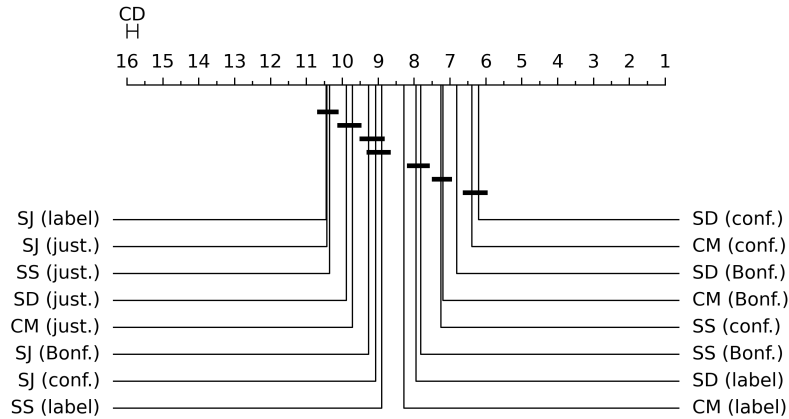


Figure 3: Significant differences for the 3B model

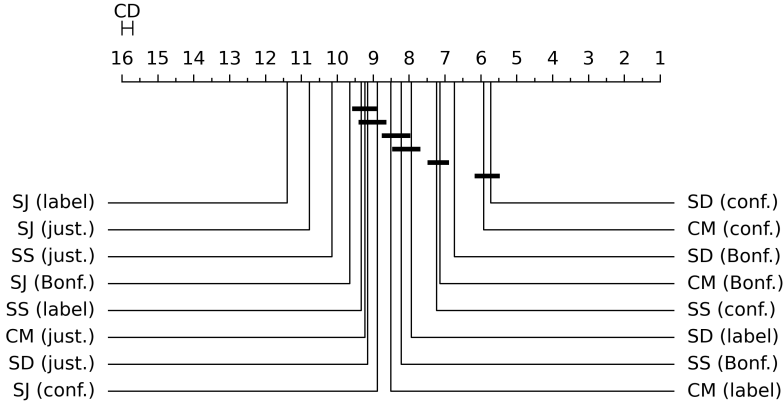


Figure 4: Significant differences for the 7B model

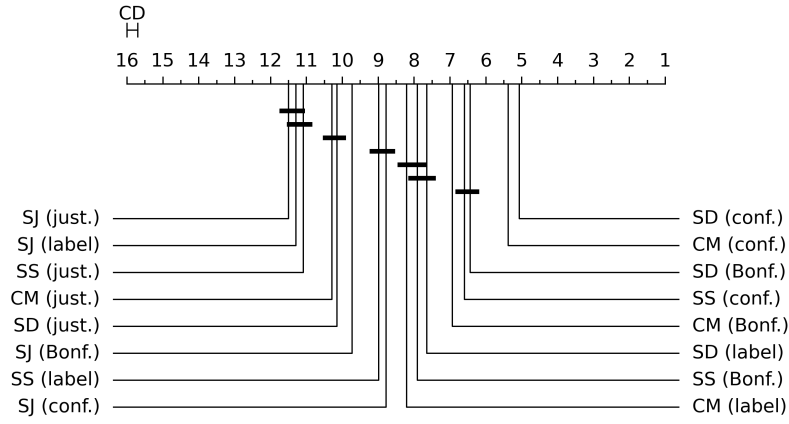


Figure 5: Significant differences for the 14B model

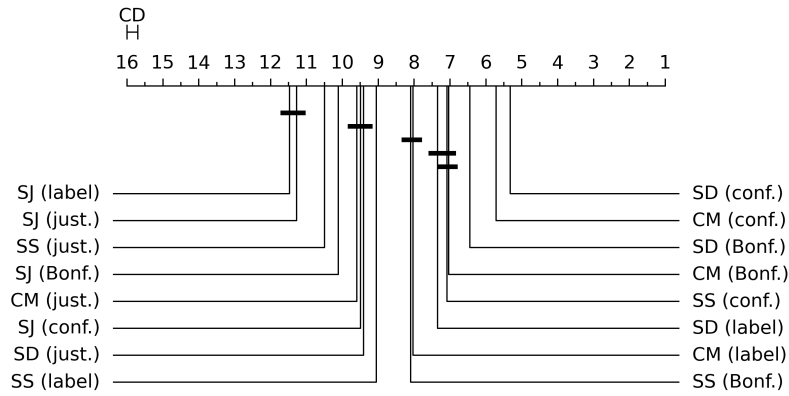


Figure 6: Significant differences for the 32B model

3.3. Summary of findings

3.3.1. RELATIVE PERFORMANCE OF ALGORITHMS

The ranking of the martingale algorithms is quite consistent across datasets, models, and signals. Simple Jumper is not competitive for any of the datasets, while the benefit of the composite martingale over Sleeper/Drifter is typically small (if any), suggesting that the added computational cost of combining the latter with Sleeper/Stayer and Simple Jumper may not be motivated.

3.3.2. RELATIVE PERFORMANCE OF SIGNALS

Confidence is the most reliable signal overall, consistently yielding the lowest detection times, when it is at all effective. The use of labels is competitive on easier datasets, but its relative performance degrades for smaller models on the more difficult ones. Combining all three signal types improves over any single signal for two of the multiclass datasets (BBC, NEWS). Justification is the most dataset-dependent signal: it fails completely for some or all models on FAKE, SPAM, and TWEETS, while it is the only effective signal for 3B on NEWS.

3.3.3. RELATIVE PERFORMANCE OF MODELS

Larger models generally detect faster, but the effect is uneven. The transition from the 3B to the 7B model is by far the most important step, representing a qualitative capability threshold on the harder datasets; on FAKE and NEWS, this means going from complete non-detection to reasonable detection, for some of the signals. Beyond 7B, improvements are incremental. The most notable exception is TWEETS, where the 7B model is substantially worse than the 3B model on the confidence-based and composite signals.

3.3.4. OBSERVATIONS ON DATASETS AND TASKS

The binary datasets SPAM and IMDB are relatively easy, with detection times falling well below the endpoint for most combinations, while FAKE is much harder for the 3B model but easier for larger models. Among the multiclass datasets, BBC is the easiest and most consistent, NEWS is harder (especially for 3B), and TWEETS is the most difficult; it is the only dataset where even the 32B model struggles to detect well before the endpoint, and also the only one where justification is completely uninformative across all models. BBC stands out as the only dataset where justification is consistently competitive with confidence for all model sizes.

4. Concluding Remarks

We have proposed conformal test martingales as an approach to detect label shift for LLMs, using output class labels, confidence assessments and justifications as proxies for the true class labels.

Results from an empirical investigation across six text classification tasks have been presented showing that label shift, as introduced synthetically by changing class priors, could indeed be detected. It was observed that larger shifts naturally lead to earlier detection and also that the effectiveness of the different proxies varied across tasks and models, although confidence was found to be clearly the most reliable proxy for binary datasets, and combining all three signals yielded the best results on two of the multiclass datasets. Justification failed completely for some or all models on three of the datasets, while it was the only effective signal for the smallest model on one of the datasets. Larger models were observed to lead to earlier detection of label shift, but the effect was not monotonic.

The ranking of the algorithms for computing the martingales in terms of detection times was observed to be quite consistent across datasets, models, and signals, with Sleeper/Drifter and the composite martingale outperforming the others, while Simple Jumper was the least competitive. This result is quite distinct from an earlier investigation (Boström, 2026), in which Sleeper/Stayer was found to be the most effective among twelve considered martingales, including Simple Jumper and Sleeper/Drifter, in a majority of investigated cases. The prominent feature of Sleeper/Drifter to extend its pool of betting functions and gradually shift the thresholds of the functions, in contrast to Sleeper/Stayer, here turned out to be crucial for effectively detecting label shift for LLMs.

There are several directions for future research. The design of prompts and scoring functions for textual output could be refined in several ways, e.g., by exploiting any available labeled texts, which can be expected to lead to more effective text-based detectors than the investigated ones, which rely on reference texts in the form of keywords generated by the LLM itself. Other output of the LLMs, such as token logits, could also provide useful information for label shift detection. The empirical investigation focused on abrupt label shift only, which could be complemented and combined with both continuous and label-conditional shift, where also the effect on relevant performance measures could be investigated. More generally, the investigated framework may be studied in contexts beyond text classification and label shift; given suitable scoring functions, drift may be detected for LLM output for several other tasks, in addition to predicting class labels.

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Appendix A: TWEETS Classification

Prompt:

Classify the following tweet into exactly one of these categories:
negative, neutral, positive

Definitions:

- negative: the tweet expresses negative sentiment (unhappiness, criticism, frustration, anger, sadness)
- neutral: the tweet is factual, ambiguous, or expresses no clear positive or negative sentiment
- positive: the tweet expresses positive sentiment (happiness, praise, excitement, gratitude, satisfaction)

Also provide:

- confidence: how certain you are that the chosen label is correct
(0.0 = completely unsure, 1.0 = completely certain)
- justification: 5-10 comma-separated keywords summarising your reasoning.

Guidelines:

- Use EXACTLY one of the categories listed above - no other values are permitted.
- If the text spans multiple categories, pick the most dominant one and use a lower confidence value.
- Confidence should reflect uncertainty; use the full range of values when appropriate.
- Confidence should roughly match how often you expect to be correct on similar examples.

MAYDAY?!

Output:

```
{
  "label": "negative",
  "confidence": 0.7,
  "justification": "urgency, distress, emergency, concern, alarm"
}
```

Appendix B: TWEETS Reference Texts (positive)

Prompt:

Generate 100 reference justification texts.

Purpose:

These texts will be used as semantic reference points when comparing model-generated justifications using cosine similarity.

Task:

Generate short justification-style entries (5-10 comma-separated keywords each) explaining why a tweet would be classified as positive sentiment.

Requirements:

- Each entry must be 5-10 comma-separated keywords.
- Focus on emotional or linguistic cues indicating positivity.
- Cover diverse kinds of positive sentiment.
- Maximize TOPIC diversity.

Typical cues include:

- expressions of happiness or joy
- praise or appreciation
- gratitude or thankfulness
- excitement or enthusiasm
- satisfaction or approval
- uplifting or encouraging language

Output:

happy, joyful, smiling, celebration, content
 praise, admiration, respect, acknowledgment, commendation
 thankful, grateful, appreciation, blessings, kindness
 ...